ARRAY (Arrays, Constructor, Parameterized constructor)

Write a java program to create an array of Student objects. Each student has the following attributes; Name(String), RegNo (String), marks of 3 subjects (double).

A class should include:

1. A parameterized constructor to initialize all attribut
2. A default constructor that assigns default value
3. A method display() to print the details of each student.

Ans:

```
class Student {

String name;
String regNo;
double mark1, mark2, mark3;

// Default Constructor

Student ( ) {
name = "Not Assigned";
regNo = "N/A";
mark1 = 0;
mark2 =0;
mark3 =0;

}
```

```
// Parameterized constructor
Student (String name, String regNo, double mark1,
        double mark2, double mark3)
 {

this.name =name;
this.regNo = regNo;
this.mark1 =mark1;
this.mark2 = mark2;
this.mark3=mark3;
 }
// Method to display student details
 void display ( ){
 System.out.println ("Student Name: " + name);
 System.out.println ("Student Reg. No: "+ regNo);
 System.out.println ("Marks");
 System.out.println ("Subject 1 :" + mark1);
System.out.println ("Subject 2 :"+mark2);
System.out.println ("Subject 3 :"+mark3);
    }
 }
```

```
public class StudentArray {

    public static void main(String[] args){

        Student[] students = new Student[3]; // Create array
```

declares an array that can store Student obj references

→ name of array

creates an array with 3 elements

// Here, we create array of student objects. of 3 student objects.

```
        // Initialize objects using both constructors.

        Students[0] = new Student(); // Default constructor

        Students[1] = new Student ("Anu", "ABC001", 90.5, 91.5, 92.5);

        Students[2] = new Student ("Kiran", "ABC002", 80.5, 81.5,
                                                        82.5);

        // Display details of all students.

        System.out.println("Student Details");

        For(int i=0; i < students.length ; i++)
          {
          students[i].display();
          }
    }
}
```